

6. In a ripple tank, circular water waves are produced by two vibrators S_1 and S_2 of the same frequency vibrating in phase. Their separation is 3.5λ , where λ is the wavelength of the waves. Figure 6.1 shows the two circular waves propagating on the water surface at a certain moment. Line L is a line connecting all points P which have path difference $S_1P - S_2P = 0$.

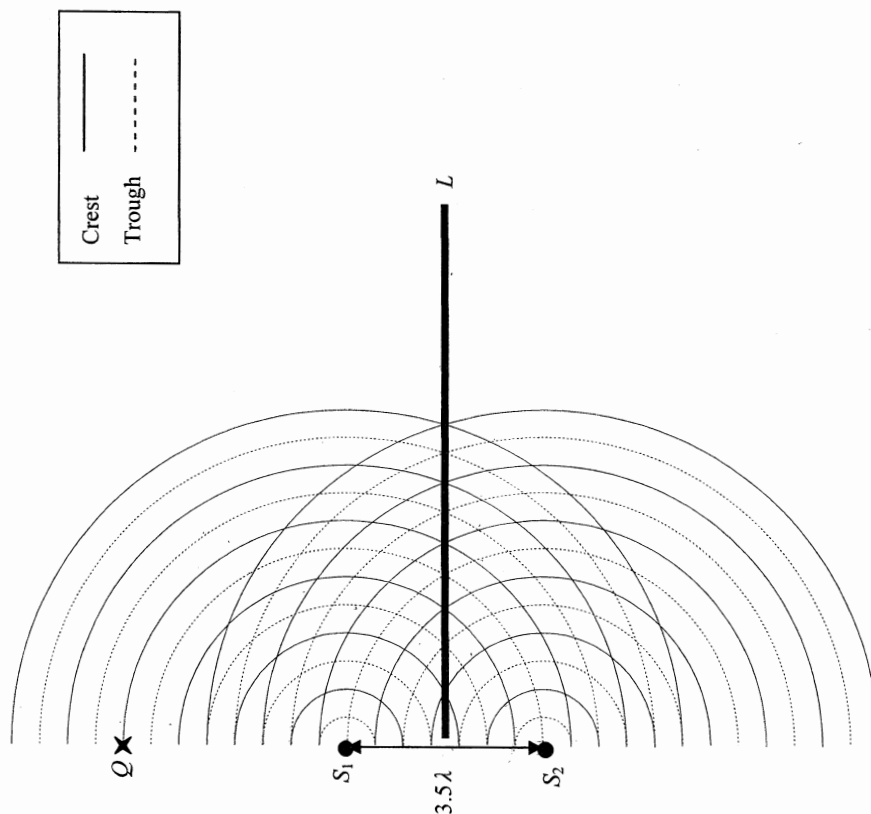


Figure 6.1

- (a) Draw and label a line in Figure 6.1 connecting all points P which have path difference

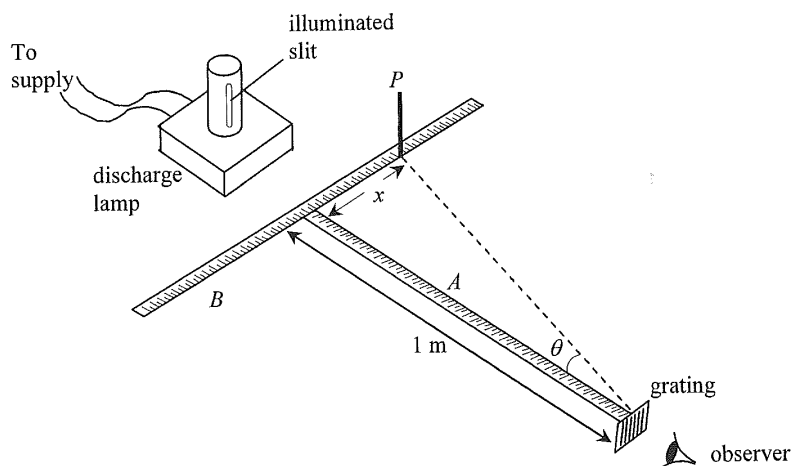
(i) $S_1P - S_2P = \lambda$ (label it as L_1)

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7. Figure 7.1 shows an experimental set-up to determine the wavelength of monochromatic light emitted from the vertical narrow slit of a discharge lamp. A , B are two mutually perpendicular metre rules on the bench with rule A pointing towards the lamp. A diffraction grating with vertical lines is placed at one end of rule A . A vertically mounted pin P is moved along rule B until the pin is in line with the diffracted image of the second-order to the observer. The corresponding distance x is measured for finding the diffraction angle θ .

Figure 7.1



The grating has 300 lines per mm and x is found to be 0.38 m for the **second-order** image.

- (a) (i) Calculate the diffraction angle θ .

(1 mark)

- *(ii) Hence find the wavelength of the light from the lamp.

(3 marks)

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- (a) (iii) Give **ONE** advantage of measuring the position of the second-order image instead of the first-order one. (1 mark)

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- (b) In the experiment, the illuminated slit may not be well aligned along metre rule *A*. Suggest one way to reduce this error. (2 marks)

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- (iii) Only five bright dots are observed on the screen such that the separation between the 1st and 5th dots is 1.56 m. Find λ . (3 marks)

- (ii) Briefly explain why the accuracy of the experiment can be improved. The double slit is now replaced by a diffraction grating with 400 lines per mm. (1 mark)

- (i) For the same set of apparatus, suggest a way to increase the average separation between the bright dots on the screen. (1 mark)
- Figure 7.1 shows a set-up for measuring the wavelength λ of light emitted by a laser pointer. Several bright dots of average separation about 2 mm can be seen on the screen.

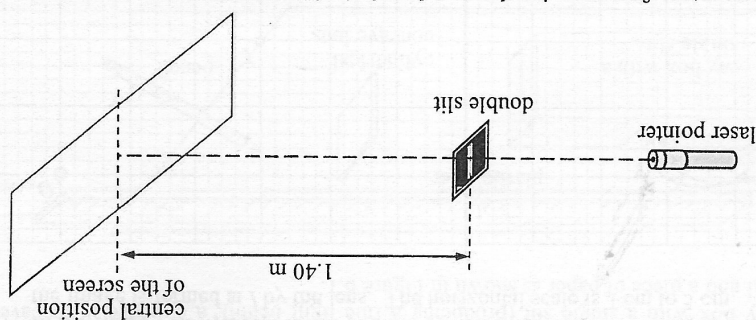
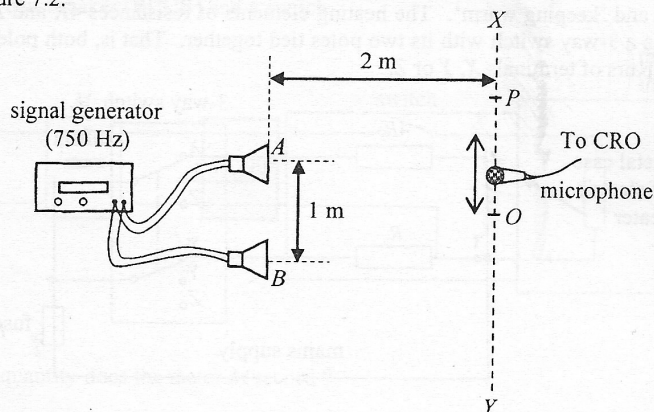


Figure 7.1

7. *(a)

- (b) To measure the speed of sound in air, a student connects two loudspeakers, A and B , to a signal generator as shown in Figure 7.2.

Figure 7.2

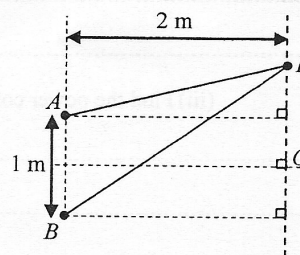


The separation of A and B is 1 m. A microphone is used to pick up the sound along the line XY at a distance of 2 m from the loudspeakers. The central maximum is at point O while the next maximum is at point P .

- (i) With reference to the above settings, use the fringe separation equation $\Delta y = \frac{\lambda D}{a}$ in double-slit interference to find the wavelength λ of sound is not accurate. Explain briefly. (1 mark)

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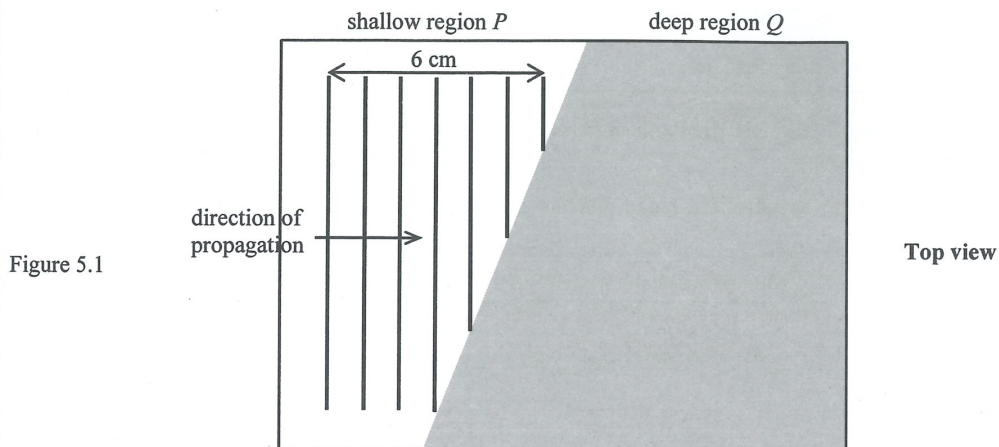
- (ii) The distance between O and P is found to be 1 m when the signal generator is set at 750 Hz. By considering the path difference $PB - PA$, use the results of the experiment to find the speed of sound in air. (3 marks)



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5. A ripple tank has a shallow region P and a deep region Q . Straight water wave of frequency 10 Hz is travelling in the shallow region as shown in Figure 5.1 when viewed from above.



- (a) The separation between seven crests in the shallow region is found to be 6 cm as shown.

- (i) Find the wavelength of the wave in the shallow region. (1 mark)

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- (ii) What is the wave speed in the shallow region? (1 mark)

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- (b) The water wave then propagates into the deep region where the wavelength of the wave is double that in the shallow region.

- (i) State the frequency of the water wave in the deep region. (1 mark)

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- (ii) On Figure 5.1, sketch the wave pattern in the deep region. (2 marks)

- (iii) Name the phenomenon occurred across the boundary and explain its cause. (2 marks)

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- *6. After removing the protective coating and the reflective layer of a compact disc (CD), the CD becomes a diffraction grating with evenly spaced parallel slits (Figure 6.1(a)). The set-up in Figure 6.1(b) employs a laser beam which can be used to measure the slit separation d .

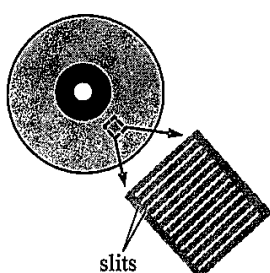


Figure 6.1(a)

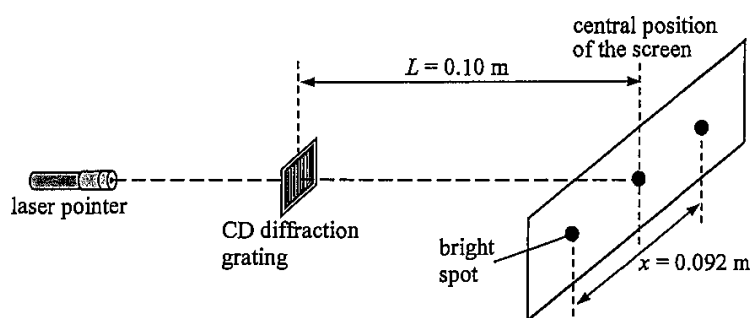


Figure 6.1(b)

Given that the separation between the CD diffraction grating and the screen, L , is 0.10 m, the separation between the first-order maxima, x , is 0.092 m and the wavelength λ of the red laser light beam used is 650 nm.

(a) Estimate d .

(3 marks)

(b) State and explain the change in x if green laser light is used.

(2 marks)

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